CLAUDE CODE PLUGIN

AI Design Harness

Find out what's known, check you're building the right thing, then design it the way your team would — with an AI agent.

github.com/sadakoa/AI-design-harness

WHY

Why it exists

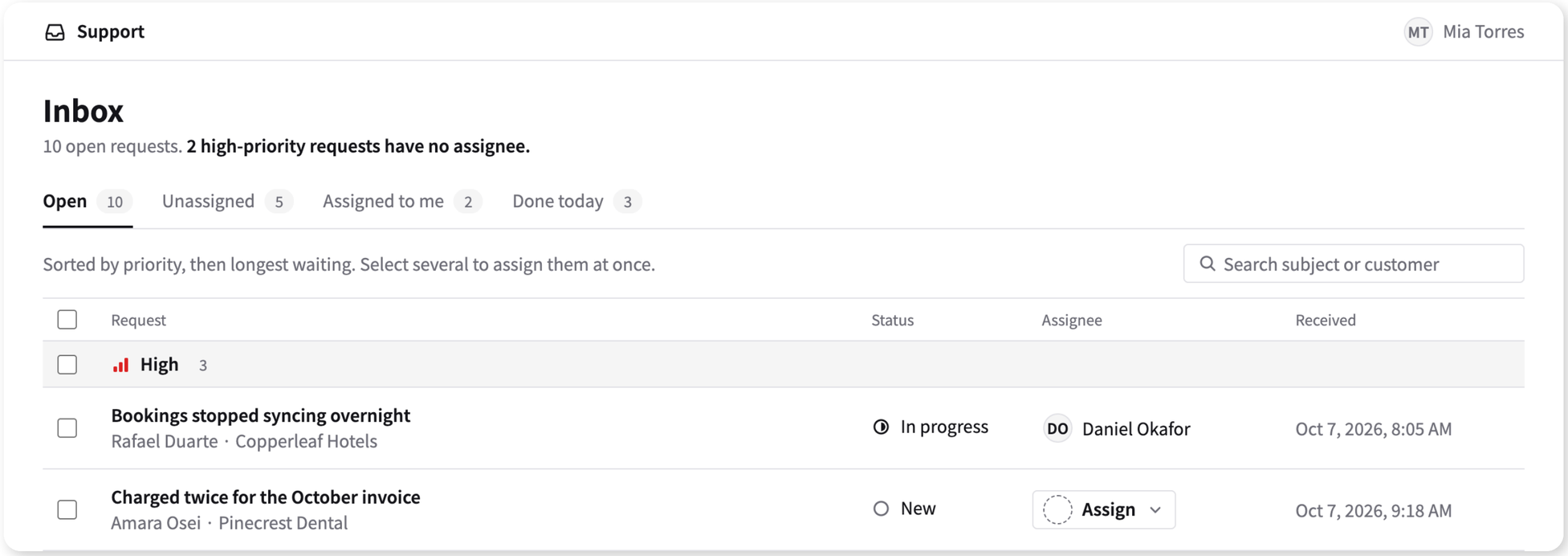
What we know about users and our design rules usually live in people's heads. The harness writes both down as short notes in your repo, so an AI agent can follow them. You just ask, in your own words.

Today	With the harness
❌ Each request starts from scratch	✅ Each request starts from what's already known
❌ PRDs aren't checked against what we know	✅ Each PRD is checked against users and jobs
❌ One long doc mixes rules and one-off calls	✅ One question, one file
❌ Review comments are opinions, not rules	✅ Findings cite a rule, or say it's taste
❌ Lessons get lost in old chat threads	✅ Lessons flow back into product and design

WHAT YOU GET

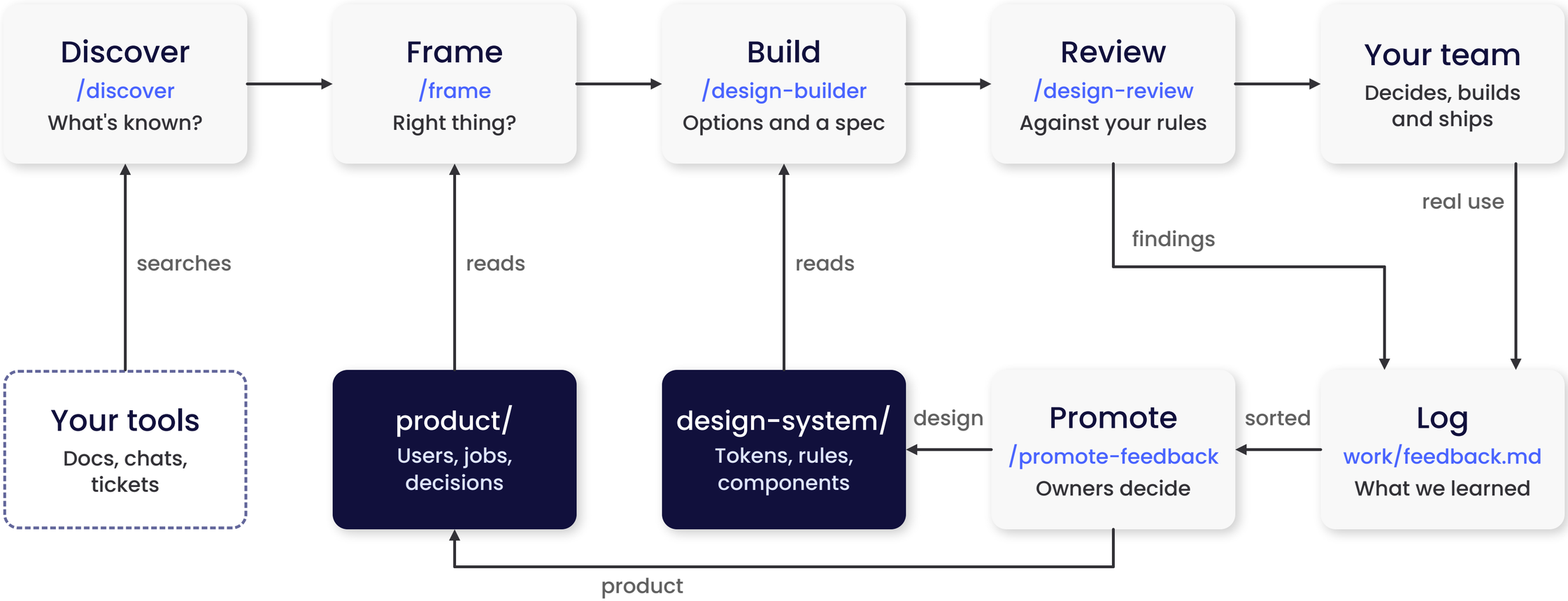
One proposal per screen or flow

A plain answer first, then about three clickable mockups, a recommendation and a spec on one HTML page, reviewed against your rules. Your team still decides, and builds it.



HOW IT FLOWS

From idea to what you learned



IDEA 1

Look before you check

Requests arrive as a PRD, a line in Slack, or a problem nobody has written up. The agent sizes the work, then looks for what's already known.

small	medium	large
A word, a small fix Nothing to look up. It goes straight to the check.	**A change to a place you have** Searches your docs, chats, tickets and numbers, in about ten minutes.	**A new screen, flow or idea** Also looks at how other products handle the same job.

Every finding has a source and says how sure it is: seen, said or guessed. No PRD? It drafts one for someone to correct.

IDEA 2

Check the problem first

Before any design, the request is checked against product/: who it's for, which job, which decisions and constraints. A go reaches design; so does a decide first, if you say go ahead.

go	decide first	stop
Support inbox For agents (U-01) who need to start with what's urgent (J-01). examples/prd/support-inbox.md	**Hand-off flow** Four screens, but no job covers its main goal yet. examples/prd/hand-off.md	**Community forum** Customer log-ins go against PD-01. Public posts break CON-02. examples/prd/community-forum.md

Every user and job in product/ is marked fact, hypothesis or open, so a PRD built on a guess gets caught.

IDEA 3

One question, one file

Each folder's INDEX.md says which file answers what, and what to read for each task.

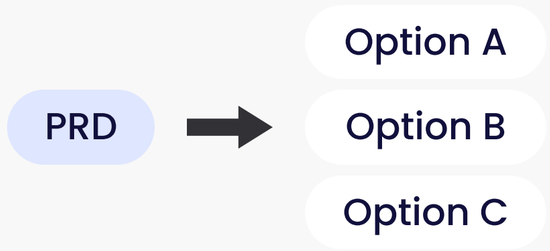
product/ — what to build and why	design-system/ — how it looks and works
Who uses the product? users.md	Which exact values do we use? tokens/
What are they trying to get done? jobs.md	What do we prioritize? principles.md
What do we call things? concepts.md	What must we never do? rules.md
What places are there? structure.md	How do we write UI text? writing.md
What limits us? constraints.md	Which component should I use? components/
What do we stand for? principles.md	How do screens and flows fit? patterns/
What have we decided? decisions.md	Why did we decide this? decisions.md
Where do the full docs live? sources.md	What do approved screens look like? screens/

Setup starts with the dark rows. The faded ones are added when a task needs them.

IDEA 4

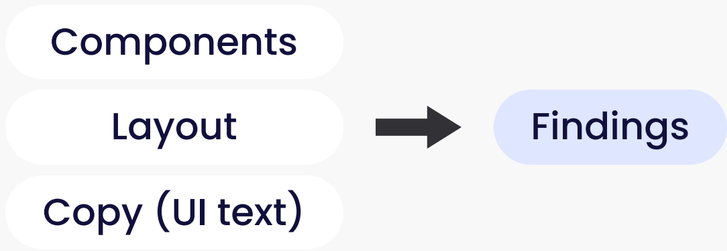
Build wide, review narrow

Build: explore options



For bigger work, five to eight rough directions come first, and you pick. Then about three options that differ in structure, built with your product's own components when it has them.

Review: one area at a time

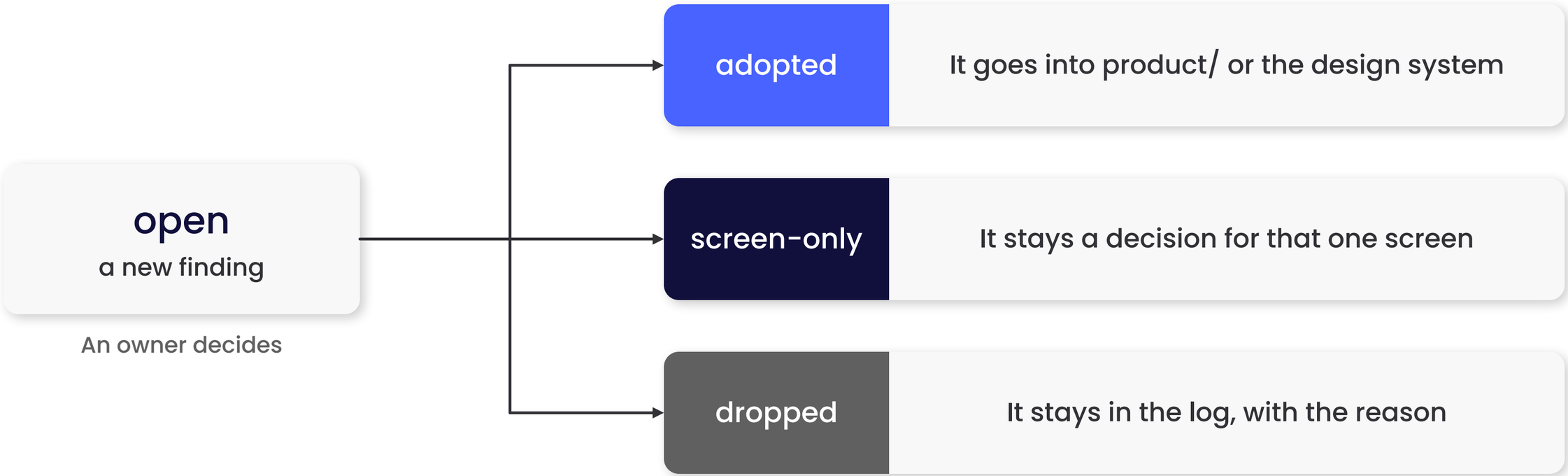


First: does it do the job? Then one area at a time. Findings cite a rule, or say it's taste.

IDEA 5

Feed what you learn back

Review findings, what real use shows, and decisions nobody wrote down go into one log, marked product (we built the wrong thing) or design (we built it the wrong way).



IN PRACTICE

Running it as a team

Owners are the people in CODEOWNERS: product owners for product/, design owners for design-system/.

First months	Then	As you go	Approvals
Only owners change product/ and the design system. Collect findings first.	Designers open PRs too. Start with UI text rules (writing.md).	Sort each feature's notes as it merges. An owner catches the rest weekly.	With two or more owners, require their approval on GitHub.

NOT FOR	
	Screens unlike anything your product has today. The agent drafts options; someone with design experience should finish them.

START

Your first hour

For whoever sets it up. You need Claude Code, Node 20 or later, and git.

1. Add the plugin in Claude Code, once per machine
2. In your product's repo, say “Set up the design harness”
3. Check its drafts, name your owners, and merge
4. Ask for one screen, in your own words
5. Open the proposal it writes, and react to it

What it adds	
product/	What to build and why
design-system/	How it looks and works
work/	Proposals and the log
scripts/	check and sync
The skills	Come with the plugin

Built on ideas from Canary's NestUI and Reiwa Travel's design-builder write-up

NEXT

Let's try it on one screen

The screen	The repo	The owner	Questions
[Screen to try]	[URL]	[Name]	[Channel]

[Name] asks the agent to design it, with a PRD or in a few words. It looks into it and checks it first. Then we look at its answer and the proposal together. Anything that could apply elsewhere goes into the log.

APPENDIX

design-builder, step by step

01
Check it
Looks into it and frames it, if nobody has yet

02
Define the problem
Whose job (U-, J-) gets better, and how

03
Ask you
Only what changes the design (up to 5)

04
Research
Approved screens first, then others

05
Explore
Rough directions for bigger work, then about three options

06
Recommend
Pick one, tie it to a job and a principle

07
Write up
Everything on one HTML page

08
Review
Checks its own pick, then one reply to you

APPENDIX

Reading by task

● always · ○ when relevant · design-system/INDEX.md. product/INDEX.md has its own: frame reads all of it.

File	Build	Components	Layout	Copy	Promote
components/inventory.md	○	●			
tokens/tokens.css	●	●	○		
tokens/tokens.json					○
foundations/principles.md	●	●	●	○	●
foundations/rules.md	●	●	●	●	●
foundations/writing.md	●			●	○
components/GUIDE.md	○	●			○
components/<name>.md	○	○			○
patterns/GUIDE.md	●		●		○
screens/README.md	●		○		
decisions.md					●

APPENDIX

What the check catches

npm run check runs on your machine and on every pull request.

File missing from an INDEX

In product/ or design-system/, or its Answers line doesn't match

Out of sync

Your code changed since the last npm run sync

Off-token CSS

Hard-coded colors or unknown CSS variables (R-01)

Duplicate IDs

Two entries share an ID: U-, J-, PD-, R-, FB- and the rest

Fact or guess?

A user or job not marked fact, hypothesis or open

Old notes

A file not reviewed in 90 days, or a log row open for 30